

GEN211 Probability and Statistics

Lecture 0: Course Outline





Lecture contents

- General information
- Course syllabus
- Assessment
- Textbooks



General Information



Contact information

- Lecturers:
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Course Syllabus



Course contents

- Lecture (1): Descriptive statistics I: Data organization and data presentation.
- Lecture (2): Descriptive statistics II: Data summarization.
- Lecture (3): Probability rules.
- Lecture (4): An introduction to correlation, interpolation, and least squares fitting.
- Lecture (5): Counting and probability with counting.
- Lecture (6): Discrete probability distributions.
- Lecture (7): Continuous probability distributions.
- Lecture (8): Confidence intervals and hypothesis testing.



Course learning outcomes (CLOs)

- CLO 1: Be able to organize, present, and summarize a given raw data set, and identify and apply the Chebyshev's theorem.
- CLO 2: Recognize the basic concepts in probability theory and differentiate between classical, empirical, and subjective probabilities.
- CLO 3: Understand the basic notions of correlation, interpolation, least squares, normal equations, and data models (periodic data and data linearization).
- CLO 4: Identify the different discrete probability distributions and learn how to evaluate the mean and standard deviation.
- CLO 5: Identify the normal, Gamma, and exponential distributions and understand the central limit theorem.
- CLO 6: Find the confidence interval for the mean and perform hypothesis testing.



Assessment



Grades distribution

- Take-home assignments (5 + 5%)
- In-class assignments (5%)
- Mini-projects (5 + 5%)
- Quizzes (7 + 8%)
- Midterm exam (20%)
- Final exam (40%): Comprehensive, i.e., covering the whole course material



Textbooks



Textbooks

- A. Bluman, Elementary Statistics: A step by step approach, 7th Edition, Mc-Graw-Hill.
- T. Sauer, Numerical Analysis, 2nd edition, Pearson.
- R. E. Walpole, R. H. Myers, S. L. Myers, and K. Ye, Probability & Statistics for Engineers & Scientists, 9th edition, Prentice Hall.
- K. H. Rosen, Discrete Mathematics and its Applications, 7th edition, McGraw-Hill.

